

환기장애와 수면 무호흡

호흡기내과 김태훈

1. 폐쇄 수면 무호흡의 진단검사는?
(5점)

- ☒ 1) 수면 다원검사
- ☐ 2) 야간 코골이 증상
- ☐ 3) 두꺼운 목의 임상 지표
- ☐ 4) 폐기능검사

2. 폐쇄 수면 무호흡 환자의 1차 치료는?
(5점)

- ☐ 1) 체중감량 "비만이 원인일 경우" 체중감량도 가능
- ☒ 2) 기도 양압기
- ☐ 3) 약물치료
- ☐ 4) 하악골 절제술

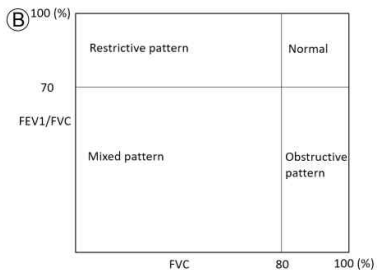
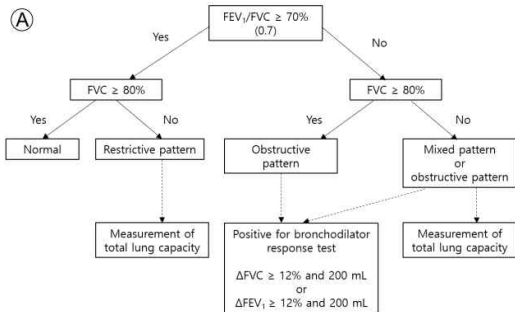
학습 목표

- 환기장애의 기전과 이상을 설명할 수 있다.
- 폐쇄성 무호흡 증후군의 임상양상을 이해하고 필요한 진단검사 및 치료를 설명할 수 있다.

환기장애란? 폐는 수동적인 장기

- 환기장애는 공기를 들이마시고 내쉬는 과정(ventilation)의 장애
 - ✓ 폐쇄성(obstructive)
 - ✓ 제한성(restrictive)
 - ✓ 혼합형(mixed)
 - ✓ 가스교환 장애(diffusion impairment)

폐기능검사



폐쇄성 환기장애

(Obstructive Ventilatory Defect)

- 주요 질환

- ✓ COPD

- ✓ 천식

- ✓ 기관지확장증

- ✓ 세기관지염(small airway disease)

- ✓ 종양/이물/기도 협착

- FEV1/FVC 감소

- 질병진행에 따라 FEV1 감소

- $\uparrow$ RV, TLC, $\uparrow$ FRC (공기 포획; Air trapping)

제한성 환기장애 (Restrictive Ventilatory Defect)

- **주요질환 (폐 vs. 폐외 [흉막, 신경 근육])**
 - **폐 실질 질환:** 간질성 폐질환
 - ✓ Interstitial lung disease; Idiopathic pulmonary fibrosis
 - ✓ Nonspecific interstitial pneumonia
 - ✓ Hypersensitivity pneumonitis
 - ✓ Organizing pneumonia
 - **흉곽/근육 문제:** 흉벽 기형, 비만, 신경근육질환
 - **흉막 질환:** 흉막 석회화 (결핵 병태, 직업성 폐질환), 흉수

Blood gas analysis

- pH
- pO₂
- pCO₂
- HCO₃

- 호흡부전
 - ✓ Type 1 = hypoxemia 저산소증
 - ✓ Type 2 = hypercapnea 저환기로 인한 high CO₂

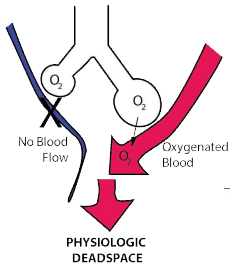
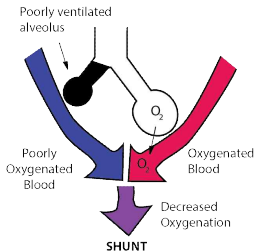
Type 1

- **$\text{PaO}_2 < 60 \text{ mmHg}$**
- **PaCO_2 정상 또는 감소 ($\leq 45 \text{ mmHg}$)**
→ 산소 공급 실패(Oxygenation Failure)
- 병태생리
- **V/Q mismatch** (가장 흔함)
 - 폐렴, 폐부종, 일측 무기폐, COPD exacerbation 일부
- **Shunt** (산소 반응이 잘 안 됨)
 - ARDS, 심한 폐렴, intracardiac shunt
- **Diffusion impairment** (산소 전달이 안됨)
 - ILD, emphysema
- **Low FiO_2 -> 산소 공급으로 해결**
 - 높은 고도, 저산소 환경

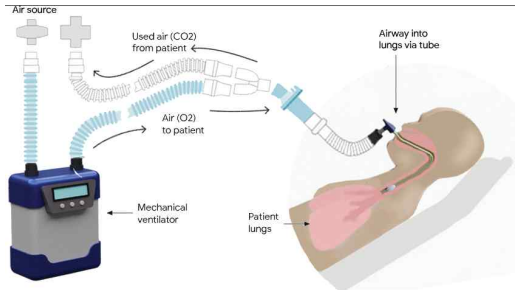
Type 2

- **$\text{PaCO}_2 > 45 \text{ mmHg}$ (+ $\text{pH} < 7.35$)**
→ 환기 실패(Ventilation Failure)
- 병태생리
- **Alveolar hypoventilation**
 - 호흡근 약화, 신경근육질환
- 기도 폐쇄/호기 저류(air trapping)
 - COPD, 천식
- 중추 호흡저하
 - 약물(마약성 진통제, 진정제)
- 흉곽 운동 제한
 - 비만(OHS), 기흉, 흉벽 기형

Shunt and hypoventilation



산소 공급해도
산소 분압 완전회복 안됨

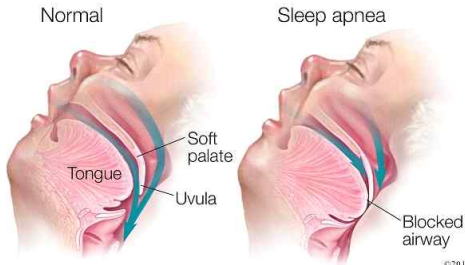


Approach to the patient with sleep disorder

수면중 환기 저하

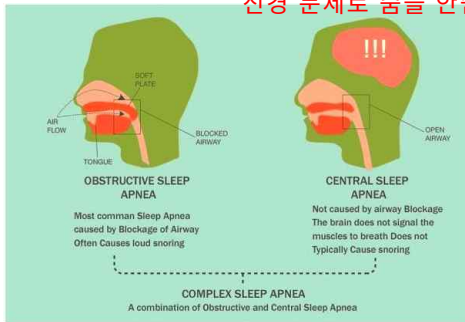
• 수면 장애의 주요 증상

- Sleepiness or tiredness during the day
- Difficulty initiating or maintaining sleep at night
- Unusual behaviors during sleep



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MAYO

신경 문제로 숨을 안쉬



공기 전달 통로 문제

호흡 신호, 호흡 중추 문제

Obstructive sleep apnea and Central sleep apnea

폐쇄 수면 무호흡

- Impaired ventilation during sleep
- Disruption of sleep

치료: 강제 환기

- Comorbidity
 - Hypertension
 - Mood disorder
 - Cognitive disorder
 - Cardiovascular disease (heart failure, atrial fibrillation)
 - Diabetes mellitus
 - Behavioral problems (child)

Definitions (OSA)

- **Obstructive sleep apnea (폐쇄 수면 무호흡; OSA)**
 - Symptoms of nocturnal breathing (코골이, 콧바람, 헐떡임, 무호흡)
 - Daytime sleepiness or fatigue
- **Sleep study(수면 다원검사; polysomnography)**
 - Five or more episodes of obstructive(or hypopnea) apnea/hour
 - Absent of symptoms if AHI ≥ 15 /hr

Risk factors

TABLE F19–1 Evaluation of Patients for OSA

High Risk Patients for OSA	Historical Information	Physical Examination
Obesity (BMI >35)	Snoring?	Increased BMI?
Congestive heart failure	Witnessed apneas or breathing pauses?	Retrognathia?
Atrial fibrillation	Gasping or choking at night?	Nasal obstruction?
Refractory hypertension	Hypertension?	Narrow oropharynx?
Type 2 diabetes	Nonrefreshing sleep?	High Mallampati Score?
Nocturnal arrhythmias	Sleepiness? (Epworth Sleepiness Scale)	Increased neck circumference?
CVA	Nocturia?	Evidence of cor pulmonale?
Pulmonary hypertension	Morning headaches?	
High-risk driving populations	Decreased concentration?	
Preoperative for bariatric surgery	Memory loss?	
	Decreased libido?	
	Irritability?	

Modified from Epstein LJ, Kristo D, Strollo PJ, et al: Clinical guideline for the evaluation, management and long-term care of obstructive sleep apnea in adults, *J Clin Sleep Med* 5:263-276, 2009.

BMI, Body mass index; CVA, cerebrovascular accident; OSA, obstructive sleep apnea.

BOX F19-1 ICSD-3 Diagnostic Criteria Obstructive Sleep Apnea: Adults

(A AND B) OR C SATISFY THE CRITERIA

- A. The presence of one or more of the following:
1. The patient complains of sleepiness, nonrestorative sleep, fatigue, or insomnia symptoms.
 2. The patient wakes with breath holding, gasping, or choking.
 3. The bed partner or other observer reports habitual snoring, breathing interruptions, or both during the patient's sleep.
 4. The patient has been diagnosed with hypertension, a mood disorder, cognitive dysfunction, coronary heart disease, stroke, congestive heart failure, atrial fibrillation, or type 2 diabetes mellitus.

- B. Polysomnography (PSG) or out-of-center sleep testing (OCST¹) demonstrates:

Five or more predominantly obstructive respiratory events (apneas, hypopneas or respiratory effort-related arousals 2 [RERAs]) 3 per hour of sleep during a PSG or per hour of monitoring (OCST¹).

OR

- C. PSG or OCST¹ demonstrates:

Fifteen or more predominantly obstructive respiratory events (apneas, hypopneas, or RERAs)³ per hour of sleep during a PSG or per hour of monitoring (OCST¹).

Adapted from reference 3.

Notes:

1. OCST may underestimate the true number of obstructive respiratory events per hour because actual sleep is not usually recorded. The term respiratory event index (REI) may be used to denote event frequency based on monitoring time rather than total sleep time.
2. Respiratory effort related arousals cannot be scored using OCST because arousals cannot be identified.
3. Respiratory events defined according the latest version of the *AASM Manual for the Scoring of Sleep and Associated Events*.

Key historical points

- 낮에 심한 졸림
- 수면 중 심한 코골이, 헐떡거림, 무호흡
- 성격 변화
- 아침에 원인이 없는 두통
- 만족스럽지 않은 수면

이학적 검사

- 비만도 (BMI)
- 혈압
- 이비인후과 진찰(귀-코-인두) Mallampati score (3 or 4)
- 두꺼운 목의 두께(Neck circumference)
- (남자 > 43 cm [17 inches], 여자 > 35cm [16 inches])

Mallampati Airway Classification



Modified Mallampati Airway Classification



Class I

Class II

Class III

Class IV

Soft palate and
entire uvula
visible

Soft palate and
part of uvula
visible

Soft palate
± base of
uvula visible

Soft palate
not visible

FIGURE F19-1 ■ Mallampati (MP) and modified Mallampati (MMP) airway classification. In the Mallampati maneuver, patients are instructed not to emit sounds but to open the mouth as wide as possible and protrude the tongue as far as possible. In the MMP, the patient is instructed to open the mouth as wide as possible without emitting sounds. (MP, Reproduced with permission from Friedman M, Tanyeri H, La Rosa M, et al: Clinical predictors of obstructive sleep apnea, *Laryngoscope* 109:1901-1907, 1999; and MMP, Reproduced with permission from Nuckton TJ, Glidden DV, Brownder WS, Claman DM: Physical examination: Mallampati score as an independent predictor of obstructive sleep apnea, *Sleep* 29:903-908, 2006.)

OSA prediction with Neck circumference

BOX F19-2 Prediction of Obstructive Sleep Apnea Based on Adjusted Neck Circumference

Neck circumference (NC)=NC measured in cm

A. If hypertension present, +4

B. If habitual snoring present, +3

C. If gasping or choking present, +3

Adjusted NC = NC + A + B + C

Probability of obstructive sleep apnea (OSA) based on adjusted NC:

<43 cm (17 inches) = low probability

43–48 cm (17–19 inches) = moderate probability

> 48 cm (19 inches) = high probability

From Flemons WW: Obstructive sleep apnea, *N Engl J Med* 347:498-504, 2002.

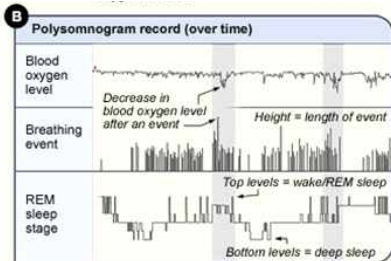
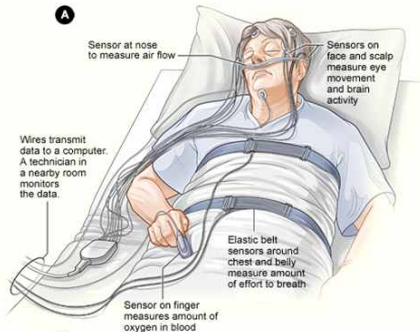
Laboratory testing

- Not indicated if disease was suspected
- Complete blood counts -> polycythemia(increased hematocrit)
- Blood gas analysis -> respiratory acidosis (hypercapnia), O₂
- Pulmonary function test (폐기능 검사) -> COPD
- Chest radiography -> underlying lung disease



¥ Diagnostic testing

- 수면 다원검사
- **Polysomnography** is gold standard test.



Polysomnography finding

BOX F19-4 Polysomnography Findings in Adult Obstructive Sleep Apnea

EEG FINDINGS

- Increased WASO and stage N1
- Reduced stage N3
- Reduced stage R (REM sleep)
- Increased respiratory arousals

RESPIRATORY FINDINGS

- Snoring
- Obstructive, mixed, and central apneas
- Obstructive hypopneas
- AHI: mild 5 to <15/hr, moderate 15–30/hr, severe >30/hr

- **Postural OSA** AHI supine $>2 \times$ AHI nonsupine
- AHI REM $>$ AHI NREM (common finding)
- Apnea duration REM $>$ NREM
- Arterial Oxygen Desaturation
 - Lowest SaO_2 during REM sleep
 - Longest REM periods in the early morning hours typically associated with the lowest SaO_2

CARDIAC

- Cyclic variation in heart rate

AHI, Apnea–hypopnea index; EEG, electroencephalogram; NREM, non–rapid eye movement; OSA, obstructive sleep apnea; REM, rapid eye movement; SaO_2 , arterial oxygen saturation; WASO, wakefulness after sleep onset.

Parameter definitions

- **AHI (Apnea-hypopnea index)**
 - Number of apneas and hypopneas per hour of sleep
 - **Apnea** (호흡기류가 <10% 10초이상 지속),
 - **Hypopnea** (호흡기류가 <30% 10초이상지속, and **desaturation**)
- **Severity**
 - Mild = AHI 5 to < 15/hr
 - Moderate = AHI 15 to <30/hr
 - Severe = AHI ≥30/hr
- **Desaturation** usually defined drop in SaO_2 >3 or 4% (or <88%)
- **RDI (Respiratory disturbance index)** 뇌파검사 필요해 사용하기 까다로움
 - **AHI** + respiratory effort-related arousals (**RERA**)
 - RERA = 10초 이상 무호흡이나 저호흡은 아니나, 호흡노력에 의한 각성이 확인된 경우

Central sleep apnea

TABLE F29-3 Summary of Diagnostic Criteria for Some of the Major CSA Syndromes

Disorder	Symptoms	PSG findings	Other Criteria
Primary central sleep apnea	+ Symptoms	1. CAHI ≥ 5 per hour (hr) 2. Number of CA+CH $>50\%$ of the total A+H	Absence of CSB No evidence of daytime or nocturnal hypoventilation
CSA with CSB	+ Symptoms OR atrial fibrillation or flutter, congestive heart failure, or a neurologic disorder	1. CAHI ≥ 5 per hour (hr) 2. Number of CA+CH $>50\%$ of the total A+H 3. 1 and 2 occur during either a diagnostic PSG or PAP titration PSG	CA or CH meet criteria for CSB
Treatment emergent CSA	+ Symptoms	Diagnostic study - predominantly obstructive events, AHI ≥ 5 /hr PAP titration study-after obstructive events have resolved shows CAHI ≥ 5 /hr and $>50\%$ of events are central apneas or central hypopneas	CSA not better explained by another CSA disorder (e.g., no CSB, no opioids)
CSA due to drug or substance	+Symptoms	1. CAHI ≥ 5 per hour (hr) 2. Number of CA+CH $>50\%$ of the total A+H 3. 1 and 2 occur during either a diagnostic PSG or PAP titration PSG	Absence of CSB • Believed to be caused by opioid or respiratory depressant • Sleep-related hypoventilation may be present
CSA due to medical disorder without CSB	+Symptoms	1. CAHI ≥ 5 per hour (hr) 2. Number of CA + CH $>50\%$ of the total A + H	Absence of CSB • Believe to be cause by medical or neurological condition • not caused by drug or substance • Sleep-related hypoventilation may be present

CA, Central apnea; CH, central hypopnea; CAHI, central apnea + central hypopnea index; CSB, Cheyne-Stokes breathing, as defined in the AASM Scoring Manual; PSG, polysomnography.

+ Symptoms=presence of sleepiness, insomnia, awakening SOB, snoring, witnessed A, apnea; H, hypopnea.

수면 다원검사

QUESTIONS

1. What type of respiratory event is shown in Figure 1

Oronasal flow
Present

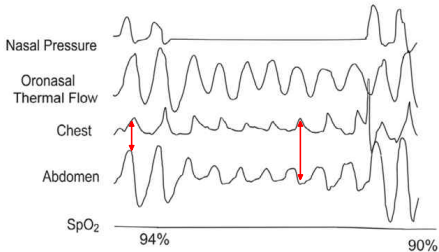
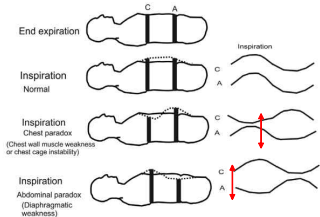
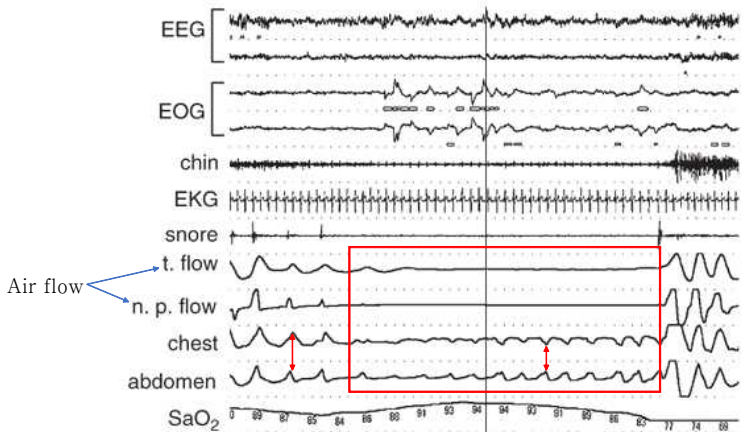


FIGURE P20-1 ■ A respiratory event.



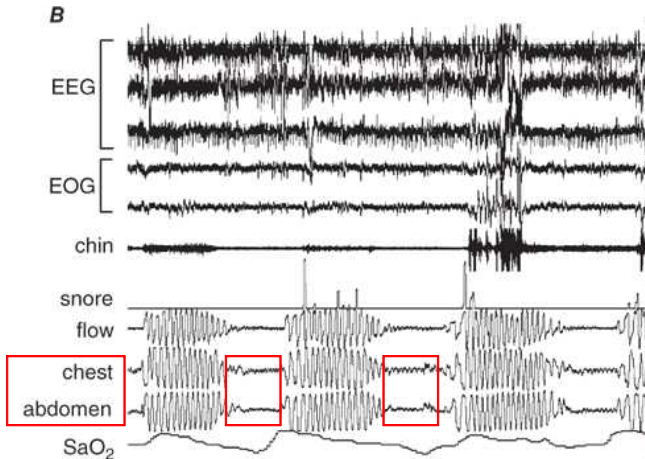
Paradox signal of chest and abdomen
(Obstructive apnea or hypopnea)

A

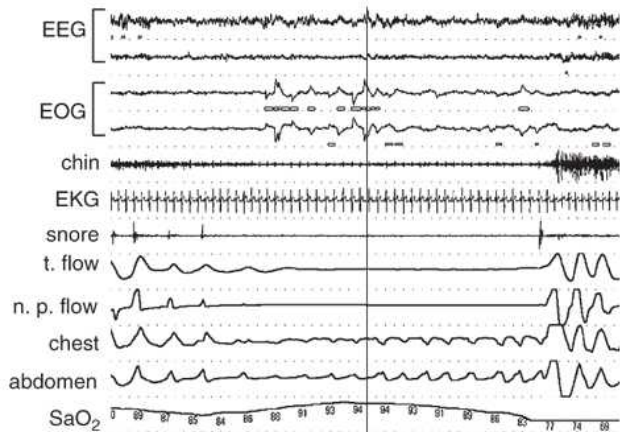


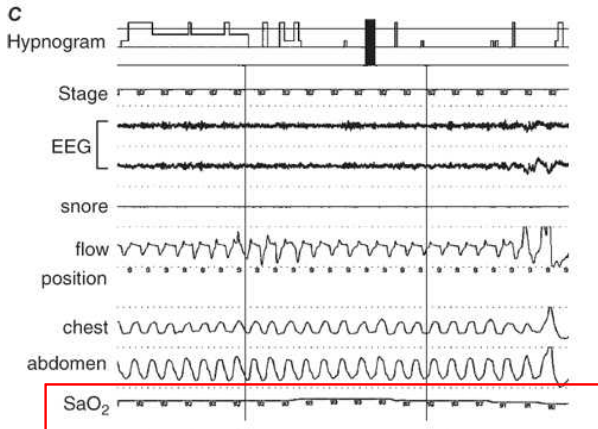
Paradox signal of chest and abdomen
(Obstructive apnea or hypopnea)

Obstructive apnea. A. There are 30 s of **no airflow**, as shown in the **nasal pressure** (n. p. flow) and thermistor-measured flow (t. flow). Note the **presence of chest-abdomen paradox**, indicating respiratory effort against an occluded airway.



Obstructive apnea. B. Central apnea in a patient with Cheyne-Stokes respiration due to congestive heart failure. **The flat chest-abdomen tracings** indicate the absence of inspiratory effort during the central apneas





Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
Copyright © McGraw Hill. All rights reserved.

Obstructive apnea. C. Hypopnea. Partial obstruction of the pharyngeal airway can limit ventilation, **leading to desaturation (a mild decrease in this patient, from 93 to 90%)** and arousal.

Treatments

BOX F22-1

American Academy of Sleep Medicine Practice Parameter Recommendations for Medical Treatment of Obstructive Sleep Apnea

WEIGHT REDUCTION

- Successful dietary weight loss may improve the AHI in obese patients with OSA. (Guideline)
- Dietary weight loss should be combined with primary treatment of OSA. (Option)
- Bariatric surgery may be adjunctive in treatment of OSA in obese patients. (Option)

POSITIONAL THERAPIES

- Positional therapy, consisting of a method that keeps the patient in a nonsupine position, is an effective secondary therapy or can be a supplement to primary therapies for OSA in patients who have a low AHI in the nonsupine versus the supine position. (Guideline)

OXYGEN SUPPLEMENTATION

- Oxygen supplementation is not recommended as a primary treatment for OSA. (Option)

NASAL CORTICOSTEROIDS

- Topical nasal corticosteroids may improve the AHI in patients with OSA and concurrent rhinitis and, thus, may be a useful adjunct to primary therapies for OSA. (Guideline)

MODAFINIL, ARMODAFINIL

- Modafinil is recommended for treatment of residual excessive sleepiness in patients with OSA, who have sleepiness despite effective PAP treatment and who are lacking any other identifiable cause for their sleepiness. (Standard)

OTHER TREATMENTS

- Protriptyline, SSRIs, aminophylline, estrogen preparations with or without progesterone, and short-acting decongestants. These treatments are NOT recommended.

Adapted from Morgenthaler TI, Kapen S, Lee-Chiong T, et al: Practice parameters for the medical therapy of obstructive sleep apnea, *Sleep* 29:1031-1035, 2006.

AHI, Apnea-hypopnea index; OSA, obstructive sleep apnea; PAP, positive airway pressure; SSRIs, selective serotonin reuptake inhibitors.

효과는 좋지만
compliance
낮음

Postural OSA

(AHI of supine > 2* AHI of non-supine)



돌아눕기 효과 좋음

FIGURE P46-1 ■ **A and B,** A lateral positioning device. The ZZsoma Positional Sleeper. (From Permut I, Diaz-Abad M, Eissam C, et al: Comparison of positional therapy to CPAP in patients with positional obstructive sleep apnea, *J Clin Sleep Med* 6:238-243, 2010.

Treatments of choice; **PAP** ★

BIPAP (Bilevel positive airway pressure)

이상기도양압



FIGURE F23-5 ■ Positive airway pressure (PAP) interface options. *Left*, A nasal pillows mask (Swift LT for her by ResMed). *Right*, A full face mask (Mirage Quattro by ResMed). (Images used with permission. EPR, ResMed, Swift and Quattro are trademarks of ResMed Limited and are registered in the U.S. Patent and Trademark Office.)

PAP의 불편으로 compliance 낮아 다른 방법 연구해봤지만 다 그닥..

How about non-PAP treatment?

- Dual-block mandibular advancement device
- HNS during sleep (설하신경 자극술)
- Maxillo-mandibular osteotomy (상하악 절골술)
- Carbonic anhydrase inhibitors (탄산 탈수효소 억제제)

PICO: should a custom-made **dual-block mandibular advancement device** or **CPAP** be used for adult patients with OSA?

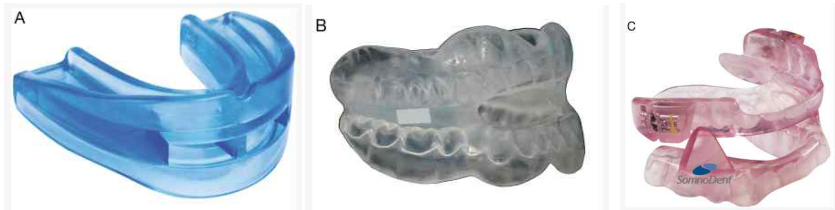


Figure 2 Examples of mandibular advancement devices. (A) “Boil and bite” device (SleepPro Easifit); (B) typical appearance of a monobloc appliance; (C) example of an adjustable bespoke device (SomnoDent Classic)

PICO : should **HNS during sleep** or **no treatment** be used for adult patients with OSA?

- We suggest that **HNS should not be used as first-line treatment** for OSA patients in general (conditional recommendation, very low quality of evidence).

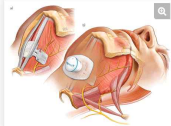
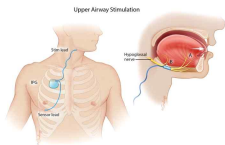


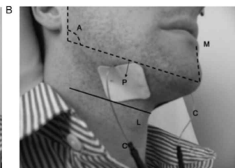
FIGURE 1
Submental musculature showing the implanted stimulator straddling the geniohyoid muscles and hypoglossal nerve branches bilaterally and to the disposable patch and activation unit. The images are for illustrational purposes only and it should be noted that the surgical anatomy might differ from person to person, thereby requiring adjustment to the specific placement of the implanted stimulator over the hypoglossal nerves.



Upper airway stimulation (UAS) through the external use of stimulation of the hypoglossal nerve and geniohyoid muscle has emerged as an effective treatment option. The device allows an implantable device in the neck through small incision with an external electrode placed around the posterior border of the hypoglossal nerve and a sensor implanted in the implanted space in the neck to detect inspiration.

턱뼈를 앞으로 빼 기도 확보/얼굴형 달라지는 부작용 있지만 효과는 확실

Bilateral hypoglossal nerve stimulation for treatment of adult obstructive sleep apnoea European Respiratory Journal 2020 55: 1901320; DOI: 10.1183/13993003.01320-2019



A, Frontal view of the placement of the patches.
B, Lateral view of the placement of the patches (P) on the skin, halfway (arrow) between midline (M) of the chin and angle of the mandible (indicated by the dotted lines) on the submental region, above a line (indicated by the solid line) cranial of the larynx (L). Patches were then connected by cables (C) to the stimulation device.

Author J. Seymour pictured with permission.

PICO : should maxillo-mandibular osteotomy or CPAP be used for adult patients with OSA?

- In adult patients with OSA, the task force **suggests using** either MMO or CPAP (conditional recommendation, very low quality of evidence).

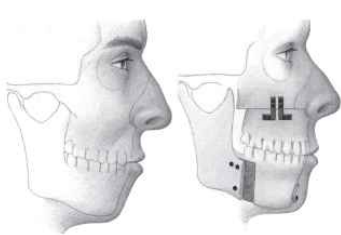


Fig. 4. Bimaxillary advancement. Ref. 10 with permission from Ilchokak.

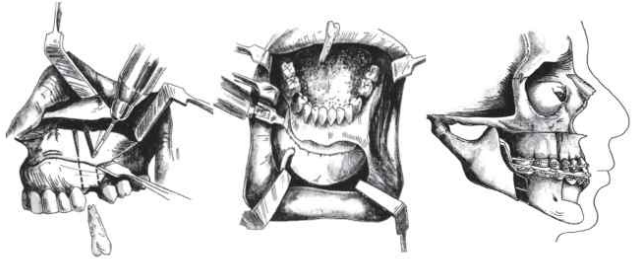


Fig. 5. Modified bimaxillary advancement. Ref. 24 with permission from John Wiley and Sons.

PICO : should **carbonic anhydrase inhibitors** (compared to placebo) be used for adult patients with OSA?

- The literature on the use of carbonic anhydrase inhibitors in OSA is limited and dominated by studies on **acetazolamide**. The dose range in available trials is wide (36–1000 mg) and treatment periods up to 3 months have been described.

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- **Task force recommendation**

- We suggest use of carbonic anhydrase inhibitors only in the context of an RCT, as **there is no drug in this category with an approved label for OSA** (conditional recommendation, low quality of evidence)

Reference

1. Richard B. Berry and Mary H. Wagner et al. Sleep Medicine Pearls. Book. Third Edition. 2014
2. Harrison's Principles of Internal Medicine, 21e Chapter 297: Sleep Apnea
3. European Respiratory Society guideline on non-CPAP therapies for obstructive sleep apnoea